Name _____ Surname _____ Date ___/___/_____

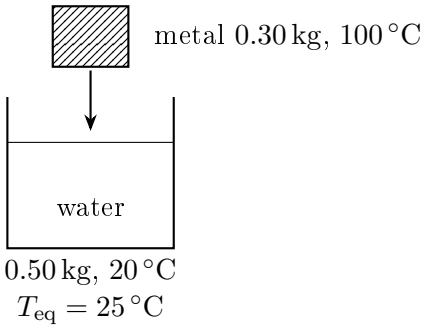
Calorimetry and mixtures: 20 exercises

Data. $c_{\text{water}} = 4200\,\text{J/(kg}\,^\circ\text{C)}$, $c_{\text{ice}} = 2100\,\text{J/(kg}\,^\circ\text{C)}$, $L_f = 3.3 \times 10^5\,\text{J/kg}$ (melting), $L_v = 2.3 \times 10^6\,\text{J/kg}$ (boiling). Unless the text says otherwise, no energy is lost to the container or to the surroundings.

Part A — Liquids and solids, no change of state

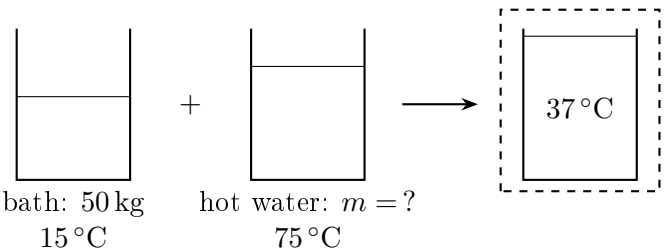
Exercise 1 — Hot metal in water

A 0.30 kg block of metal at 100 °C is dropped into 0.50 kg of water at 20 °C. The equilibrium temperature is 25 °C. Find the specific heat capacity of the metal. Which metal might it be? (Typical values: aluminium about 900, iron about 450, copper about 390, lead about 130 J/(kg °C).)



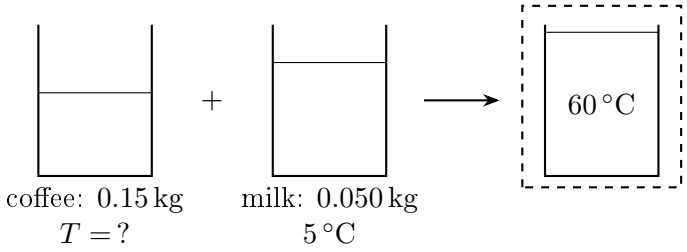
Exercise 2 — A bath

A bath contains 50 kg of water at 15 °C. How much water at 75 °C must be added to bring the bath to 37 °C?



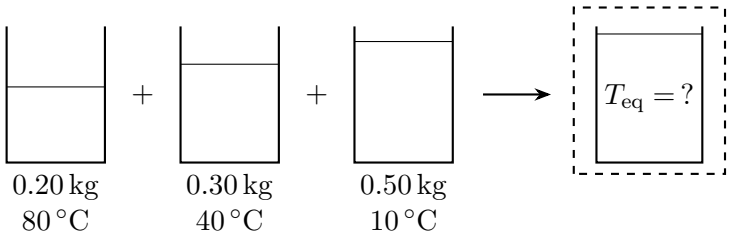
Exercise 3 — Coffee and milk

0.15 kg of hot coffee (treat it as water) is poured into 0.050 kg of cold milk (treat it as water) at 5 °C. The mixture ends at 60 °C. What was the temperature of the coffee?



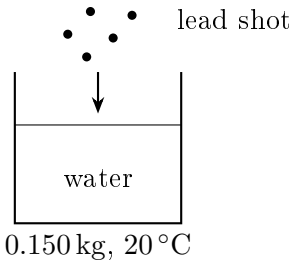
Exercise 4 — Three cups

Three amounts of water are mixed in an insulated jug: 0.20 kg at 80 °C, 0.30 kg at 40 °C and 0.50 kg at 10 °C. What is the equilibrium temperature?



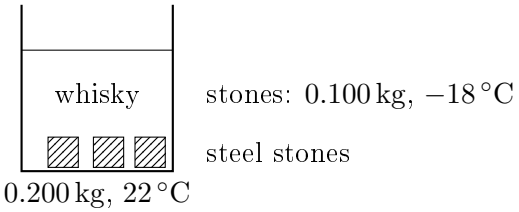
Exercise 5 — Lead shot in water

0.200 kg of lead shot at 90 °C is poured into 0.150 kg of water at 20 °C in an insulated cup (ignore the cup). The specific heat capacity of lead is 130 J/(kg °C). Find the equilibrium temperature. The lead cools by about 67 °C, the water warms by only about 3 °C: explain why.



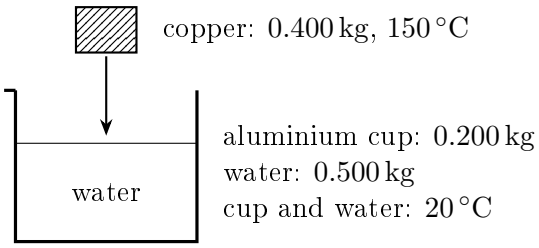
Exercise 6 — Whisky stones

0.100 kg of steel "whisky stones" ($c = 450\,\mathrm{J/(kg\,°C)}$) are taken from a freezer at $-18\,°\mathrm{C}$ and put in 0.200 kg of whisky (treat it as water) at 22 °C. Find the equilibrium temperature. Does the drink get much colder? Why, or why not?



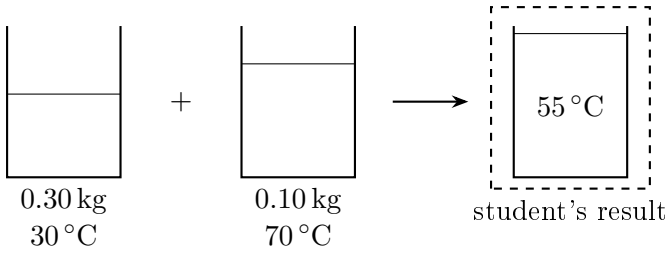
Exercise 7 — A calorimeter

An aluminium cup of 0.200 kg ($c = 900\,\mathrm{J/(kg\,°C)}$) contains 0.500 kg of water, and cup and water are both at 20 °C. A 0.400 kg block of copper ($c = 390\,\mathrm{J/(kg\,°C)}$) at 150 °C is dropped in. Find the equilibrium temperature. Three bodies: write three terms.



Exercise 8 — Find the mistake

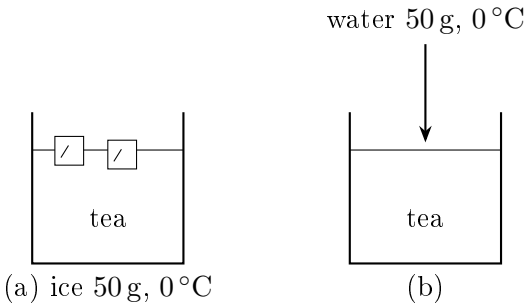
A student mixes 0.30 kg of water at 30 °C with 0.10 kg of water at 70 °C and finds an equilibrium temperature of 55 °C. Without doing the full calculation, explain why this cannot be right. Then find the correct value.



Part B — Ice and steam: which ending?

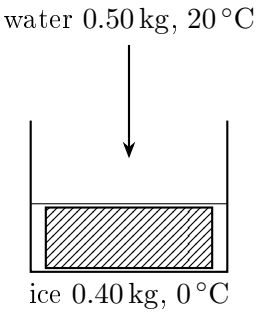
Exercise 9 — Tea with ice, tea with water

0.20 kg of tea at 80 °C (treat it as water). Compare adding 50 g of ice at 0 °C with adding 50 g of water at 0 °C: what is the equilibrium temperature in each case?



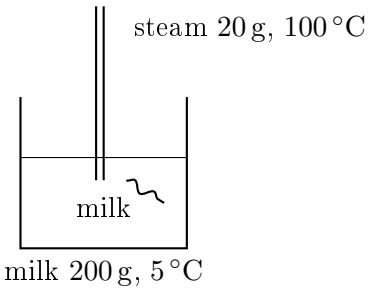
Exercise 10 — Does it all melt?

0.50 kg of water at 20 °C is poured onto 0.40 kg of ice at 0 °C. What is in the container at the end, and at what temperature?



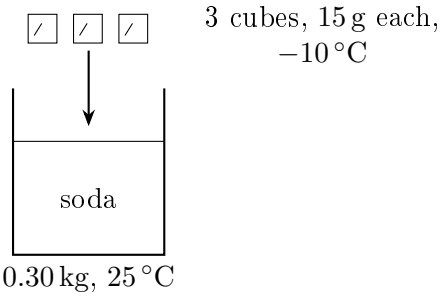
Exercise 11 — Steam as a heater

A coffee machine blows 20 g of steam at 100 °C into 200 g of milk at 5 °C. Treat the milk as water. What is the equilibrium temperature?



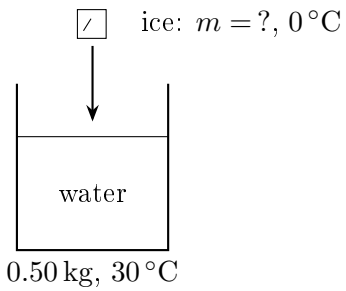
Exercise 12 — Ice cubes in a soda

Three ice cubes of 15 g each, taken from a freezer at −10 °C, are dropped into 0.30 kg of soda (treat it as water) at 25 °C. Find the equilibrium temperature. Start with the budget.



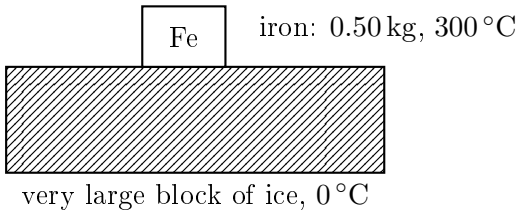
Exercise 13 — How much ice?

What mass of ice at 0 °C must be added to 0.50 kg of water at 30 °C to cool it to 10 °C?



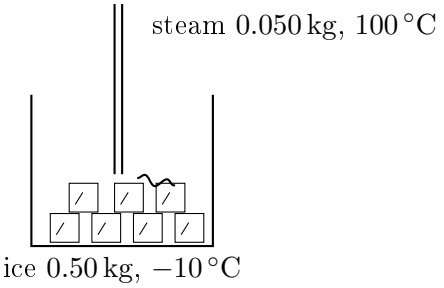
Exercise 14 — Hot iron on a block of ice

A 0.50 kg piece of iron ($c = 450\,\mathrm{J/(kg\,°C)}$) at 300 °C is placed on a very large block of ice at 0 °C. It cools down to 0 °C. What mass of ice melts?



Exercise 15 — Steam on ice

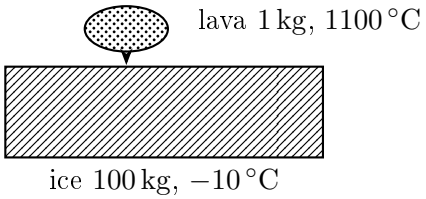
0.050 kg of steam at 100 °C is blown into 0.50 kg of ice at −10 °C in an insulated container. What is in the container at the end, and at what temperature?



Part C — Lava, laboratory and design

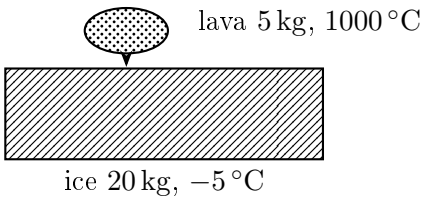
Exercise 16 — Lava on a block of ice

1 kg of lava at 1100 °C ($c \approx 1000\,\mathrm{J/(kg\,°C)}$) lands on 100 kg of ice at −10 °C. Contact is perfect and no energy escapes; ignore the solidification of the lava. Find the equilibrium temperature and how much ice melts. Start with the budget.



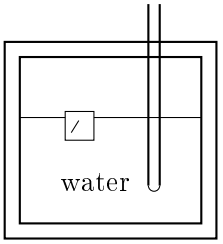
Exercise 17 — A bigger lava flow

5 kg of lava at 1000 °C ($c \approx 1000\,\mathrm{J/(kg\,°C)}$) lands on 20 kg of ice at −5 °C. What is left at the end, and at what temperature? Compare with the previous exercise: what changed, and why?



Exercise 18 — Measuring the latent heat of fusion

In an experiment, 0.020 kg of ice at 0 °C is added to 0.100 kg of water at 40 °C in a well-insulated container. After stirring, the final temperature is 20.2 °C. (a) Find the specific latent heat of fusion of ice. (b) The ice had been taken out of a bowl and was wet: its surface was covered with liquid water. Is the value you measure too large or too small? Explain.



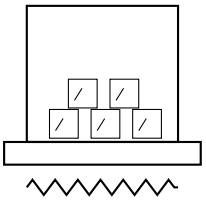
insulated container

thermometer

water 0.100 kg, 40 °C
ice 0.020 kg, 0 °C

Exercise 19 — Heating ice, step by step

0.20 kg of ice at −20 °C is heated by a 500 W heater that loses no energy to the surroundings. Find the duration of each of these four stages: (a) from −20 °C to 0 °C; (b) melting all the ice; (c) from 0 °C to 100 °C; (d) boiling away all the water. Then sketch the graph of temperature against time, to scale, and label the four stages.

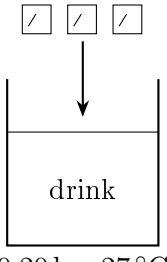


ice 0.20 kg, −20 °C

heater, 500 W

Exercise 20 — Designing the drink

0.20 kg of drink at 27 °C must be cooled with ice taken from a freezer at −18 °C. (a) What mass of ice brings the drink to exactly 5 °C, with all the ice melted? How many cubes of 20 g is that? (b) With exactly 3 cubes (60 g), what is the equilibrium temperature?



ice, −18 °C

0.20 kg, 27 °C

Solutions

Il metodo, passo per passo

1 · Il principio. In un recinto isolato nessuna energia entra o esce: quella che un corpo perde la riceve un altro. Perciò la somma di tutti i calori scambiati è zero:

$$Q_1 + Q_2 + \cdots = 0.$$

Si scrive un termine per ogni corpo che cambia temperatura o cambia stato. Non serve decidere prima chi cede e chi assorbe: lo decidono i segni.

2 · Un corpo che cambia temperatura. Per ogni corpo

$$Q = m\,c\,(T_{\text{eq}} - T_{\text{iniziale}}),$$

sempre *finale meno iniziale*, mai il contrario. Il segno viene da solo: se il corpo si scalda Q è positivo, se si raffredda è negativo. Non si mettono valori assoluti e non si scambiano le temperature per "rendere positivo" un termine. Se l'incognita è una temperatura iniziale (esercizio 3) la regola non cambia: si scrive $(T_{\text{finale}} - T)$ con T incognita. Se due corpi sono fatti della stessa sostanza (per esempio acqua) c si semplifica, ma solo dopo averlo scritto.

3 · Un corpo che cambia stato. Durante il cambiamento di stato la temperatura resta costante e il calore è $Q = +m\,L_f$ quando il ghiaccio fonde (energia assorbita) e $Q = -m\,L_v$ quando il vapore condensa (energia ceduta). Dopo il cambiamento l'acqua ottenuta è un corpo come gli altri: parte da 0 °C (dopo la fusione) o da 100 °C (dopo la condensazione) e arriva a T_{eq}, quindi ha anche il suo termine $m\,c\,(T_{\text{eq}} - T_{\text{iniziale}})$.

4 · Con ghiaccio o vapore, prima il bilancio. Non si sa in partenza se il ghiaccio fonde tutto, e scrivere l'equazione sbagliata dà un numero sbagliato senza avvisare. Si calcolano allora due quantità, entrambe riferite a 0 °C:

- Q_{caldo} (negativo): quanto cede il corpo caldo raffreddandosi fino a 0 °C;
- Q_{ghiaccio} (positivo): quanto serve al ghiaccio per scaldarsi fino a 0 °C (se parte sotto zero) e per fondere completamente.

Il segno della somma e il confronto con il solo riscaldamento del ghiaccio dicono quale dei tre finali è quello giusto:

Condizione	**Finale**	**Incognita**
$Q_{\text{caldo}} + Q_{\text{ghiaccio}} < 0$	tutto il ghiaccio fonde, $T_{\text{eq}} > 0$ °C	T_{eq}, con l'equazione completa
somma > 0, ma $\lvert Q_{\text{caldo}}\rvert$ basta a portare il ghiaccio a 0 °C	fonde solo una parte, $T_{\text{eq}} = 0$ °C	la massa fusa x
$\lvert Q_{\text{caldo}}\rvert$ non basta nemmeno a portare il ghiaccio a 0 °C	non fonde nulla, $T_{\text{eq}} < 0$ °C	T_{eq}, senza termine di fusione

Con il vapore il ragionamento è lo stesso, a 100 °C: la condensazione completa del vapore (negativa) contro il riscaldamento del corpo freddo fino a 100 °C (positivo). Se il vapore agisce sul ghiaccio (esercizio 15) il bilancio si fa a 0 °C, contando la condensazione e il raffreddamento dell'acqua fino a 0 °C.

5 · Controllo finale. T_{eq} deve cadere nell'intervallo che il finale permette: tra la più bassa e la più alta delle temperature iniziali; sopra $0\,^\circ\text{C}$ se tutto il ghiaccio fonde; sotto $0\,^\circ\text{C}$ se non fonde nulla. Se esce fuori, il finale scelto era sbagliato: si torna al bilancio. Un secondo controllo di buon senso: il corpo con $m\,c$ più grande cambia meno temperatura, e la temperatura finale gli resta vicina.

Exercise 1

Solution — The metal cools from $100\,^\circ\text{C}$ to $25\,^\circ\text{C}$, the water warms from $20\,^\circ\text{C}$ to $25\,^\circ\text{C}$. Two terms:

$$\begin{aligned} 0.30\,c\,(25-100)+0.50\cdot 4200\,(25-20)&=0\\ -22.5\,c+10\,500&=0\\ c=\frac{10\,500}{22.5}&\approx 467\,\text{J/(kg }^\circ\text{C)}. \end{aligned}$$

The first term is negative (the metal cools), the second positive (the water warms), so c comes out positive. The value is close to iron or steel.

Exercise 2

Solution — The bath warms from $15\,^\circ\text{C}$ to $37\,^\circ\text{C}$; the added water, of unknown mass m, cools from $75\,^\circ\text{C}$ to $37\,^\circ\text{C}$. Both are water, so c cancels:

$$\begin{aligned} 50\cdot 4200\,(37-15)+m\cdot 4200\,(37-75)&=0\\ 50\cdot 22-38\,m&=0\\ m=\frac{1100}{38}&\approx 28.9\,\text{kg}. \end{aligned}$$

Exercise 3

Solution — The coffee cools from the unknown T to $60\,^\circ\text{C}$; the milk warms from $5\,^\circ\text{C}$ to $60\,^\circ\text{C}$. Here the unknown is an initial temperature, so the factor is $(60-T)$:

$$\begin{aligned} 0.15\cdot 4200\,(60-T)+0.050\cdot 4200\,(60-5)&=0\\ 0.15\,(60-T)+0.050\cdot 55&=0\\ 9-0.15\,T+2.75&=0\\ T=\frac{11.75}{0.15}&\approx 78.3\,^\circ\text{C}. \end{aligned}$$

Exercise 4

Solution — Three bodies, three terms (all water, so c cancels):

$$\begin{aligned} 0.20\,(T_{\text{eq}}-80)+0.30\,(T_{\text{eq}}-40)+0.50\,(T_{\text{eq}}-10)&=0\\ T_{\text{eq}}-16-12-5&=0\\ T_{\text{eq}}&=33\,^\circ\text{C}. \end{aligned}$$

Check: the result lies between the coldest ($10\,^\circ\text{C}$) and the hottest ($80\,^\circ\text{C}$) water.

Exercise 5

Solution — The lead cools from 90 °C, the water warms from 20 °C:

$$\begin{aligned} 0.200 \cdot 130\,(T_{\text{eq}} - 90) + 0.150 \cdot 4200\,(T_{\text{eq}} - 20) &= 0 \\ 26\,(T_{\text{eq}} - 90) + 630\,(T_{\text{eq}} - 20) &= 0 \\ 656\,T_{\text{eq}} &= 2340 + 12\,600 = 14\,940 \\ T_{\text{eq}} &\approx 22.8\,^{\circ}\text{C}. \end{aligned}$$

The lead changes by 67.2 °C, the water by only 2.8 °C. The two terms have the same size, $mc\,\Delta T$, so a small change of temperature goes with a large mc: the water has $mc = 630\,\text{J/°C}$ against 26 J/°C for the lead, 24 times more. The equilibrium temperature ends up close to the temperature of the body with the larger mc.

Exercise 6

Solution — The steel warms from −18 °C, the whisky cools from 22 °C:

$$\begin{aligned} 0.100 \cdot 450\,(T_{\text{eq}} + 18) + 0.200 \cdot 4200\,(T_{\text{eq}} - 22) &= 0 \\ 45\,T_{\text{eq}} + 810 + 840\,T_{\text{eq}} - 18\,480 &= 0 \\ 885\,T_{\text{eq}} &= 17\,670 \\ T_{\text{eq}} &\approx 20.0\,^{\circ}\text{C}. \end{aligned}$$

The whisky cools by only 2 °C. The stones have a small mc (45 J/°C against 840 J/°C for the whisky) and nothing melts: they cannot take much energy. Ice cools much more because melting needs L_f, a large energy.

Exercise 7

Solution — Three bodies at the end at the same temperature: the cup (aluminium) and the water warm from 20 °C, the copper cools from 150 °C:

$$\begin{aligned} 0.200 \cdot 900\,(T_{\text{eq}} - 20) + 0.500 \cdot 4200\,(T_{\text{eq}} - 20) + 0.400 \cdot 390\,(T_{\text{eq}} - 150) &= 0 \\ 180\,(T_{\text{eq}} - 20) + 2100\,(T_{\text{eq}} - 20) + 156\,(T_{\text{eq}} - 150) &= 0 \\ 2436\,T_{\text{eq}} - 45\,600 - 23\,400 &= 0 \\ T_{\text{eq}} = \frac{69\,000}{2436} &\approx 28.3\,^{\circ}\text{C}. \end{aligned}$$

Exercise 8

Solution — *Why it cannot be right.* There is three times more cold water than hot water, so the result must lie closer to 30 °C than to 70 °C, that is below the midpoint 50 °C. A result of 55 °C is above it.

Correct value. Both are water, so c cancels:

$$\begin{aligned} 0.30\,(T_{\text{eq}} - 30) + 0.10\,(T_{\text{eq}} - 70) &= 0 \\ 0.40\,T_{\text{eq}} &= 9 + 7 = 16 \\ T_{\text{eq}} &= 40\,^{\circ}\text{C}. \end{aligned}$$

Exercise 9

Solution — **(a) With ice.** Budget first. If the tea cooled down to 0 °C: $Q_{\text{tea}} = 0.20 \cdot 4200 \cdot (0 - 80) = -67\,200\,\text{J}$. If all the ice melted: $Q_{\text{melt}} = +0.050 \cdot 3.3 \times 10^5 = 16\,500\,\text{J}$ (the ice starts at 0 °C, no warming term). The sum is $-50\,700\,\text{J}$, negative: all the ice melts and $T_{\text{eq}} > 0\,°\text{C}$. Three terms: the tea cools, the ice melts, the meltwater warms up from 0 °C to T_{eq}:

$$\begin{aligned} 0.20 \cdot 4200\,(T_{\text{eq}} - 80) + 0.050 \cdot 3.3 \times 10^5 + 0.050 \cdot 4200\,(T_{\text{eq}} - 0) &= 0 \\ 840\,T_{\text{eq}} - 67\,200 + 16\,500 + 210\,T_{\text{eq}} &= 0 \\ 1050\,T_{\text{eq}} &= 50\,700 \\ T_{\text{eq}} &\approx 48.3\,°\text{C}. \end{aligned}$$

Check: $0\,°\text{C} < 48.3\,°\text{C} < 80\,°\text{C}$.

(b) With water. The tea goes from 80 °C to T_{eq}, the cold water from 0 °C to T_{eq}:

$$\begin{aligned} 0.20 \cdot 4200\,(T_{\text{eq}} - 80) + 0.050 \cdot 4200\,(T_{\text{eq}} - 0) &= 0 \\ 0.25\,T_{\text{eq}} &= 16 \\ T_{\text{eq}} &= 64\,°\text{C}. \end{aligned}$$

The difference, 48.3 °C against 64 °C, is the energy of melting.

Exercise 10

Solution — Budget first. The water cooling down to 0 °C: $Q_{\text{water}} = 0.50 \cdot 4200 \cdot (0 - 20) = -42\,000\,\text{J}$. All the ice melting (it is already at 0 °C): $Q_{\text{melt}} = +0.40 \cdot 3.3 \times 10^5 = 132\,000\,\text{J}$. The sum, 90 000 J, is positive: the water cannot melt all the ice. Part of it melts, ice and water stay together, $T_{\text{eq}} = 0\,°\text{C}$. With x the melted mass:

$$\begin{aligned} 0.50 \cdot 4200\,(0 - 20) + x \cdot 3.3 \times 10^5 &= 0 \\ -42\,000 + 3.3 \times 10^5\,x &= 0 \\ x &= \frac{42\,000}{3.3 \times 10^5} \approx 0.127\,\text{kg}. \end{aligned}$$

At the end: ice $0.40 - 0.127 = 0.273\,\text{kg}$, water $0.50 + 0.127 = 0.627\,\text{kg}$, all at 0 °C.

Exercise 11

Solution — Budget first. The steam only condensing: $Q_{\text{cond}} = -0.020 \cdot 2.3 \times 10^6 = -46\,000\,\text{J}$. The milk warming up to 100 °C: $Q_{\text{milk}} = 0.20 \cdot 4200 \cdot (100 - 5) = 79\,800\,\text{J}$. The sum, 33 800 J, is positive: the steam cannot even bring the milk to 100 °C. So $T_{\text{eq}} < 100\,°\text{C}$ and all the steam condenses. Three terms: the steam condenses, the condensed water cools from 100 °C to T_{eq}, the milk warms from 5 °C to T_{eq}:

$$\begin{aligned} &- 0.020 \cdot 2.3 \times 10^6 + 0.020 \cdot 4200\,(T_{\text{eq}} - 100) \\ &\qquad + 0.20 \cdot 4200\,(T_{\text{eq}} - 5) = 0 \end{aligned}$$

$$\begin{aligned} -46\,000 + 84\,T_{\text{eq}} - 8400 + 840\,T_{\text{eq}} - 4200 &= 0 \\ 924\,T_{\text{eq}} &= 58\,600 \\ T_{\text{eq}} &\approx 63.4\,°\text{C}. \end{aligned}$$

Check: $5\,°\text{C} < 63.4\,°\text{C} < 100\,°\text{C}$.

Exercise 12

Solution — The ice has mass $3{\cdot}0.015 = 0.045\,\text{kg}$. Budget first. The soda cooling down to $0\,^\circ\text{C}$: $Q_{\text{soda}} = 0.30{\cdot}4200{\cdot}(0{-}25) = -31\,500\,\text{J}$. The ice warming up to $0\,^\circ\text{C}$: $0.045{\cdot}2100{\cdot}(0{+}10) = 945\,\text{J}$; then melting: $0.045 \cdot 3.3 \times 10^5 = 14\,850\,\text{J}$. The sum, $-31\,500 + 945 + 14\,850 = -15\,705\,\text{J}$, is negative: all the ice melts and $T_{\text{eq}} > 0\,^\circ\text{C}$. Four terms: the soda cools, the ice warms to $0\,^\circ\text{C}$, melts, and the meltwater warms to T_{eq}:

$$0.30 \cdot 4200\,(T_{\text{eq}} - 25) + 0.045 \cdot 2100\,(0 + 10) + 0.045 \cdot 3.3 \times 10^5 + 0.045 \cdot 4200\,(T_{\text{eq}} - 0) = 0$$

$$\begin{aligned} 1260\,T_{\text{eq}} - 31\,500 + 945 + 14\,850 + 189\,T_{\text{eq}} &= 0 \\ 1449\,T_{\text{eq}} &= 15\,705 \\ T_{\text{eq}} &\approx 10.8\,^\circ\text{C}. \end{aligned}$$

Exercise 13

Solution — The water cools from $30\,^\circ\text{C}$ to $10\,^\circ\text{C}$. The ice (already at $0\,^\circ\text{C}$) melts and the meltwater warms from $0\,^\circ\text{C}$ to $10\,^\circ\text{C}$. With m the mass of ice:

$$\begin{aligned} 0.50 \cdot 4200\,(10 - 30) + m \cdot 3.3 \times 10^5 + m \cdot 4200\,(10 - 0) &= 0 \\ -42\,000 + 3.3 \times 10^5\,m + 42\,000\,m &= 0 \\ m = \frac{42\,000}{3.72 \times 10^5} &\approx 0.113\,\text{kg} = 113\,\text{g}. \end{aligned}$$

Exercise 14

Solution — The block is very large and the iron ends at $0\,^\circ\text{C}$, so the ice stays at $0\,^\circ\text{C}$: it does not warm up. Only the iron cools, and a mass m of ice melts:

$$\begin{aligned} 0.50 \cdot 450\,(0 - 300) + m \cdot 3.3 \times 10^5 &= 0 \\ -67\,500 + 3.3 \times 10^5\,m &= 0 \\ m = \frac{67\,500}{3.3 \times 10^5} &\approx 0.205\,\text{kg} = 205\,\text{g}. \end{aligned}$$

Exercise 15

Solution — Budget first, at $0\,^\circ\text{C}$. The steam condensing and the condensed water cooling down to $0\,^\circ\text{C}$:

$$Q_{\text{steam}} = -0.050 \cdot 2.3 \times 10^6 + 0.050 \cdot 4200\,(0 - 100) = -115\,000 - 21\,000 = -136\,000\,\text{J}.$$

The ice warming up to $0\,^\circ\text{C}$: $0.50{\cdot}2100{\cdot}(0{+}10) = 10\,500\,\text{J}$; melting completely: $0.50{\cdot}3.3{\times}10^5 = 165\,000\,\text{J}$. The sum is $-136\,000 + 10\,500 + 165\,000 = 39\,500\,\text{J}$, positive: the steam cannot melt all the ice. The steam gives more than $10\,500\,\text{J}$, so the ice does reach $0\,^\circ\text{C}$: ending 2, $T_{\text{eq}} = 0\,^\circ\text{C}$, mass x melted:

$$\begin{aligned} -115\,000 - 21\,000 + 10\,500 + x \cdot 3.3 \times 10^5 &= 0 \\ x = \frac{125\,500}{3.3 \times 10^5} &\approx 0.380\,\text{kg}. \end{aligned}$$

At the end: ice $0.50 - 0.380 = 0.120\,\text{kg}$, water $0.050 + 0.380 = 0.430\,\text{kg}$, all at $0\,^\circ\text{C}$.

Exercise 16

Solution — Budget first. The lava cooling down to $0\,^\circ\text{C}$: $Q_{\text{lava}} = 1 \cdot 1000 \cdot (0 - 1100) = -1.1 \times 10^6\,\text{J}$. The ice warming up to $0\,^\circ\text{C}$: $Q = 100 \cdot 2100 \cdot (0 + 10) = 2.1 \times 10^6\,\text{J}$. The lava does not even give enough to bring the ice to $0\,^\circ\text{C}$: it is ending 3, nothing melts and $T_{\text{eq}} < 0\,^\circ\text{C}$. Two terms:

$$\begin{aligned} 1 \cdot 1000\,(T_{\text{eq}} - 1100) + 100 \cdot 2100\,(T_{\text{eq}} + 10) &= 0 \\ 1000\,T_{\text{eq}} - 1.1 \times 10^6 + 210\,000\,T_{\text{eq}} + 2.1 \times 10^6 &= 0 \\ 211\,000\,T_{\text{eq}} &= -1.0 \times 10^6 \\ T_{\text{eq}} &\approx -4.7\,^\circ\text{C}. \end{aligned}$$

Check: $-10\,^\circ\text{C} < -4.7\,^\circ\text{C} < 1100\,^\circ\text{C}$, and below $0\,^\circ\text{C}$ as the ending requires. The ice melts nothing because the mass of ice is a hundred times the mass of the lava, and the ice has $c = 2100$.

Exercise 17

Solution — Budget first. The lava cooling down to $0\,^\circ\text{C}$: $Q_{\text{lava}} = 5 \cdot 1000 \cdot (0 - 1000) = -5 \times 10^6\,\text{J}$. The ice warming up to $0\,^\circ\text{C}$: $20 \cdot 2100 \cdot (0 + 5) = 2.1 \times 10^5\,\text{J}$; melting completely: $20 \cdot 3.3 \times 10^5 = 6.6 \times 10^6\,\text{J}$. The sum is $-5 \times 10^6 + 2.1 \times 10^5 + 6.6 \times 10^6 = 1.81 \times 10^6\,\text{J}$, positive: not all the ice melts. The lava gives much more than $2.1 \times 10^5\,\text{J}$, so the ice reaches $0\,^\circ\text{C}$: ending 2, $T_{\text{eq}} = 0\,^\circ\text{C}$, mass x melted:

$$\begin{aligned} 5 \cdot 1000\,(0 - 1000) + 20 \cdot 2100\,(0 + 5) + x \cdot 3.3 \times 10^5 &= 0 \\ -5 \times 10^6 + 2.1 \times 10^5 + 3.3 \times 10^5\,x &= 0 \\ x &= \frac{4.79 \times 10^6}{3.3 \times 10^5} \approx 14.5\,\text{kg}. \end{aligned}$$

At the end: $14.5\,\text{kg}$ of water and $5.5\,\text{kg}$ of ice, with the lava, all at $0\,^\circ\text{C}$.

Comparison. In Exercise 16 the lava was too little for the ice and the temperature went to $-4.7\,^\circ\text{C}$; here the lava is five times larger and the ice five times smaller, so the lava has enough energy to bring the ice to $0\,^\circ\text{C}$ and melt most of it, but not all.

Exercise 18

Solution — **(a)** The warm water cools from $40\,^\circ\text{C}$ to $20.2\,^\circ\text{C}$; the ice melts and the meltwater warms from $0\,^\circ\text{C}$ to $20.2\,^\circ\text{C}$. With L_f unknown:

$$\begin{aligned} 0.100 \cdot 4200\,(20.2 - 40) + 0.020\,L_f + 0.020 \cdot 4200\,(20.2 - 0) &= 0 \\ -8316 + 0.020\,L_f + 1696.8 &= 0 \\ L_f &= \frac{6619.2}{0.020} \approx 3.31 \times 10^5\ \text{J/kg}. \end{aligned}$$

(b) The wet ice contains some liquid water at $0\,^\circ\text{C}$ that does not need to melt. The real mass of ice m_{i} is less than $0.020\,\text{kg}$, but the calculation uses $0.020\,\text{kg}$ in the term mL_f. The energy

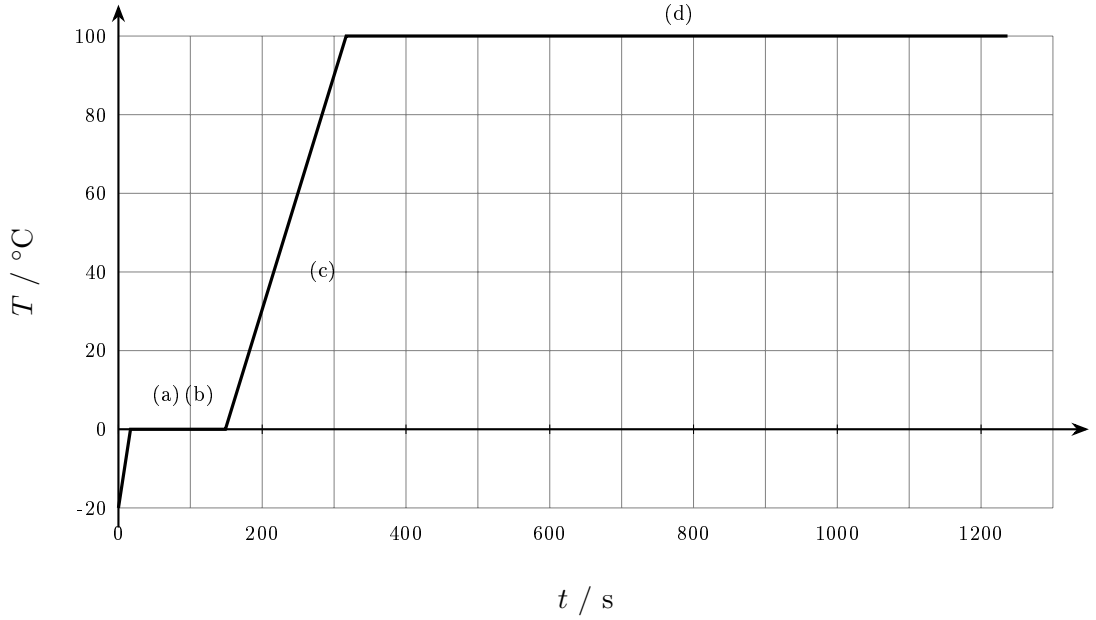
that melts the ice is the same, 6619.2 J, so $L_f = 6619.2\,\mathrm{J}/m_\mathrm{i}$ is larger than the value computed with 0.020 kg: the measured value is *too small*.

Exercise 19

Solution — The heater gives $Q = P\,t$, so $t = Q/P$ for each stage. Here Q is the energy that the ice (then the water) needs.

Stage	**Energy needed**	**Duration**	**Time since start**
(a) ice from −20 °C to 0 °C	$0.20 \cdot 2100 \cdot 20 = 8400\,\mathrm{J}$	16.8 s	16.8 s
(b) melting	$0.20 \cdot 3.3 \times 10^5 = 66\,000\,\mathrm{J}$	132 s	148.8 s
(c) water from 0 °C to 100 °C	$0.20 \cdot 4200 \cdot 100 = 84\,000\,\mathrm{J}$	168 s	316.8 s
(d) boiling	$0.20 \cdot 2.3 \times 10^6 = 460\,000\,\mathrm{J}$	920 s	1236.8 s

Total time: about 1240 s (≈ 20.6 min). Boiling takes almost three quarters of the whole time.



Exercise 20

Solution — **(a)** The drink cools from 27 °C to 5 °C. With m the mass of ice: it warms from −18 °C to 0 °C, melts, and the meltwater warms from 0 °C to 5 °C:

$$0.20 \cdot 4200\,(5 - 27) + m \cdot 2100\,(0 + 18) + m \cdot 3.3 \times 10^5 + m \cdot 4200\,(5 - 0) = 0$$

$$-18\,480 + 37\,800\,m + 330\,000\,m + 21\,000\,m = 0$$

$$m = \frac{18\,480}{388\,800} \approx 0.0475\,\mathrm{kg} = 47.5\,\mathrm{g}.$$

That is 47.5/20 ≈ 2.4 cubes: in practice 2 cubes are not enough, 3 slightly too many.

(b) Budget with $m = 0.060\,\mathrm{kg}$. The drink cooling down to $0\,^\circ\mathrm{C}$: $0.20{\cdot}4200{\cdot}(0{-}27) = -22\,680\,\mathrm{J}$. The ice warming to $0\,^\circ\mathrm{C}$: $0.060 \cdot 2100 \cdot 18 = 2268\,\mathrm{J}$; melting: $0.060 \cdot 3.3 \times 10^5 = 19\,800\,\mathrm{J}$. The sum is $-612\,\mathrm{J}$, negative but only just: all the ice melts and T_{eq} is just above $0\,^\circ\mathrm{C}$:

$$\begin{aligned} 0.20 \cdot 4200\,(T_{\mathrm{eq}} - 27) + 0.060 \cdot 2100 \cdot 18 + 0.060 \cdot 3.3 \times 10^5 + 0.060 \cdot 4200\,T_{\mathrm{eq}} &= 0 \\ 840\,T_{\mathrm{eq}} - 22\,680 + 2268 + 19\,800 + 252\,T_{\mathrm{eq}} &= 0 \\ 1092\,T_{\mathrm{eq}} &= 612 \\ T_{\mathrm{eq}} &\approx 0.56\,^\circ\mathrm{C}. \end{aligned}$$

Three cubes give an almost frozen drink; $47.5\,\mathrm{g}$ would give $5\,^\circ\mathrm{C}$.